SANSKAR ADHIKARI

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EDUCATION

Western Kentucky University — Bowling Green, KY
M.S., Geoscience — *Aug 2025 – Present*
Thesis research: Compound hydroclimatic extremes (precipitation–streamflow) in Kentucky's Salt River Basin and the role of antecedent soil moisture in extreme hydrological response
Advisor: Zachary J. Suriano, Ph.D.

Tribhuvan University, Institute of Engineering, Pulchowk Campus — Lalitpur, Nepal
B.E., Civil Engineering — *Nov 2019 – Mar 2024*
Thesis: Design of Dual Water Supply System, Gitanagar, Nagarkot, Nepal
Advisor: Shukra Raj Paudel, M.Sc.

RESEARCH INTERESTS

Hydroclimatology; extreme precipitation and flooding; compound hydroclimatic events; antecedent soil moisture and rainfall–runoff interactions; rain-on-snow processes; climate change impacts on water resources

PUBLICATIONS

Peer-Reviewed Journal Articles

Poudel, U., **Adhikari, S.**, Karki, S., & Regmi, R. K. (2025). Climate change and hydropower resilience: A CMIP6–SWAT analysis of the Kankai River Basin's energy landscape. *H2Open Journal, 8*(6), 465–490. https://doi.org/10.2166/h2oj.2025.027

Manuscripts Under Review

Suriano, Z. J., **Adhikari, S.**, & Dixon, R. D. (under review). Turbulent heat fluxes are key to understanding extreme Northern Hemispheric rain-on-snow events. *npj Hydrosphere.*

Suriano, Z. J., Hall, J., & **Adhikari, S.** (under review). On the frequency of snow and ice damming potential in the Mid-South, United States. *Cold Regions Science and Technology.*

Pinkrah, N. O., **Adhikari, S.**, Tobbe, P., & Suriano, Z. J. (under review). A comparison of Atlas 14 to precipitation frequency statistics derived from the Kentucky Mesonet. *Journal of Applied and Service Climatology.*

Manuscripts in Preparation

Adhikari, S., & Suriano, Z. J. (in preparation). Role of antecedent soil moisture conditions during compound hydroclimatic extremes: Rainfall and streamflow events in the Salt River Basin, Kentucky.

Adhikari, S., Polk, J., Fan, X., & Suriano, Z. J. (in preparation). Non-stationary peak-over-threshold study of changes in extreme rainfall events in the Salt River Basin, Kentucky.

CONFERENCE PRESENTATIONS

"Analysis of extreme hydroclimatic events: Rainfall and streamflow in the Salt River Basin, KY." *Sep 2026*
Poster presentation, EPSCoR SuperCollider, Lexington, KY.

"Role of energy fluxes on rain-on-snow events across the Northern Hemisphere." *Nov 2025*
Oral presentation, Kentucky Academy of Science Conference, University of Louisville, KY.

GRANTS AND AWARDS

Graduate Student Research/Travel Grant *2026*

Office of Research & Creative Activity, Western Kentucky University
Awarded to present M.S. thesis research on the Salt River Basin at the American Meteorological Society Annual Meeting, Denver, CO (January 2027).

RESEARCH EXPERIENCE

Climate Research Lab, Western Kentucky University Bowling Green, KY
Graduate Research Assistant *Aug 2025 – Present*

Spatiotemporal Analysis of Extreme Precipitation, Salt River Basin, Kentucky (M.S. thesis)

- Applied a nonstationary Peak-over-Threshold (POT) framework with seasonally varying, percentile-based thresholds (adapted from Acero et al., 2011) to characterize extreme precipitation using 4-km PRISM daily data (1981–2024).
- Developed reproducible Python workflows for extreme rainfall event identification and Ferro–Segers declustering, with spatial coverage criteria to quantify independent event frequency, magnitude, seasonality, and spatial variability.
- Evaluated trends using Mann–Kendall tests and Sen's slope estimation, identifying a significant increase in annual extreme rainfall frequency (+1.82 events/decade; $p < 0.001$).

Northern Hemisphere Rain-on-Snow (ROS) Events and Surface Energy Fluxes

- Conducted a hemispheric-scale analysis of ROS events using hourly ERA5 reanalysis data (1980–2025), characterizing event frequency, spatial variability, and sensitivity to precipitation and snowmelt intensity thresholds.
- Analyzed spatiotemporal variability in turbulent and radiative energy fluxes during ROS events.
- Processed ~2.5 TB of NetCDF data in documented Python/xarray and Jupyter workflows (land masking, de-accumulation, event classification, climatologies, elevation-band analysis, trend diagnostics, and geospatial visualization); published the code and documentation on GitHub.

Precipitation-Frequency Analysis and Evaluation of NOAA Atlas 14, Kentucky Mesonet (collaborative)

- Collaborated with graduate and undergraduate researchers under faculty supervision to develop precipitation-frequency statistics and intensity–duration–frequency (IDF) estimates from 5-minute observations at 66 Kentucky Mesonet stations (2015–2024).
- Analyzed precipitation extremes for durations of 5 minutes to 24 hours and compared station-based IDF estimates with NOAA Atlas 14 using six objectively matched station pairs.
- Evaluated duration-dependent differences in precipitation-frequency estimates and their implications for hydrologic design, flood-risk assessment, and regional climate services.

Tribhuvan University, Institute of Engineering, Pulchowk Campus Lalitpur, Nepal
Undergraduate Researcher

Climate Change Impacts on Hydropower Generation, Kankai River Basin, Nepal

- Assessed climate change impacts on streamflow through 2100 with a multidisciplinary team, applying a calibrated and validated SWAT model forced by bias-corrected projections from six CMIP6 global climate models under SSP2-4.5 and SSP5-8.5.
- Projected energy generation for three run-of-river hydropower projects, identifying near-term production declines and longer-term increases; results published in H2Open Journal (2025).

Undergraduate Thesis: Design of Dual Water Supply System, Gitanagar, Nagarkot, Nepal (2024)

- Led a six-member team designing a dual (pumping and gravity-fed) water supply system for a 20-year design period, including field surveys, population projections, and water demand assessment.
- Designed intake, treatment, and storage units; modeled the distribution network in WaterGEMS, prepared drawings in AutoCAD and ArcGIS, and estimated project costs (~NPR 43.1 million).

TEACHING EXPERIENCE

Western Kentucky University Bowling Green, KY
Graduate Teaching Assistant, METR 121: Introduction to Meteorology *Aug 2025 – May 2026*

- Supported undergraduates in in-person and virtual (Zoom) laboratory sessions, reinforcing core atmospheric science concepts through applied exercises and guided problem-solving.
- Assisted students with laboratory procedures, Blackboard-based assignments, concept checks, and quizzes, and identified potential grading discrepancies.

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, xarray, GeoPandas, SciPy, Dask, Matplotlib, Cartopy); R; Jupyter; Git and GitHub

Climate & Hydrologic Data: ERA5/ERA5-Land; PRISM; CMIP6; USGS NWIS; Kentucky Mesonet; NetCDF processing

Modeling & Statistics: SWAT; HEC-HMS; Peak-over-Threshold extreme value analysis; IDF analysis; Mann–Kendall and Sen's slope; time-series statistics

GIS & Engineering: ArcGIS Pro; QGIS; WaterGEMS; AutoCAD; CROPWAT

PROFESSIONAL REGISTRATION

Registered Civil Engineer, Nepal Engineering Council (NEC), Nepal *2024*